



SEA-BIRD
SCIENTIFIC

SBE37-SMP MicroCAT

Instrument Configuration

Instrument Serial Number: 37-16238
Instrument Firmware Version: 4.1
Zero Conductivity Frequency: 2831.22
Communications Format: RS232
Communications Settings: 9600 baud, 8 Data Bits, No Parity

Installed Devices/Sensors

<i>Data Format</i>	<i>Measurement</i>	<i>Sensor Type</i>	<i>Serial Number</i>	<i>Rating</i>
Count	Temperature	Internal	N/A	N/A
Frequency	Conductivity	Internal	N/A	N/A
Count	Pressure Sensor	Druck	10755345	2000m(2000 dBar)

Maximum Depth: **2000m**

CAUTION - The maximum deployment depth will be limited by the measurement range of the pressure sensor, if installed, an attached sensor, if installed, or the housing.



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SENSOR SERIAL NUMBER: 16238
CALIBRATION DATE: 22-Oct-17

SBE 37 V2 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

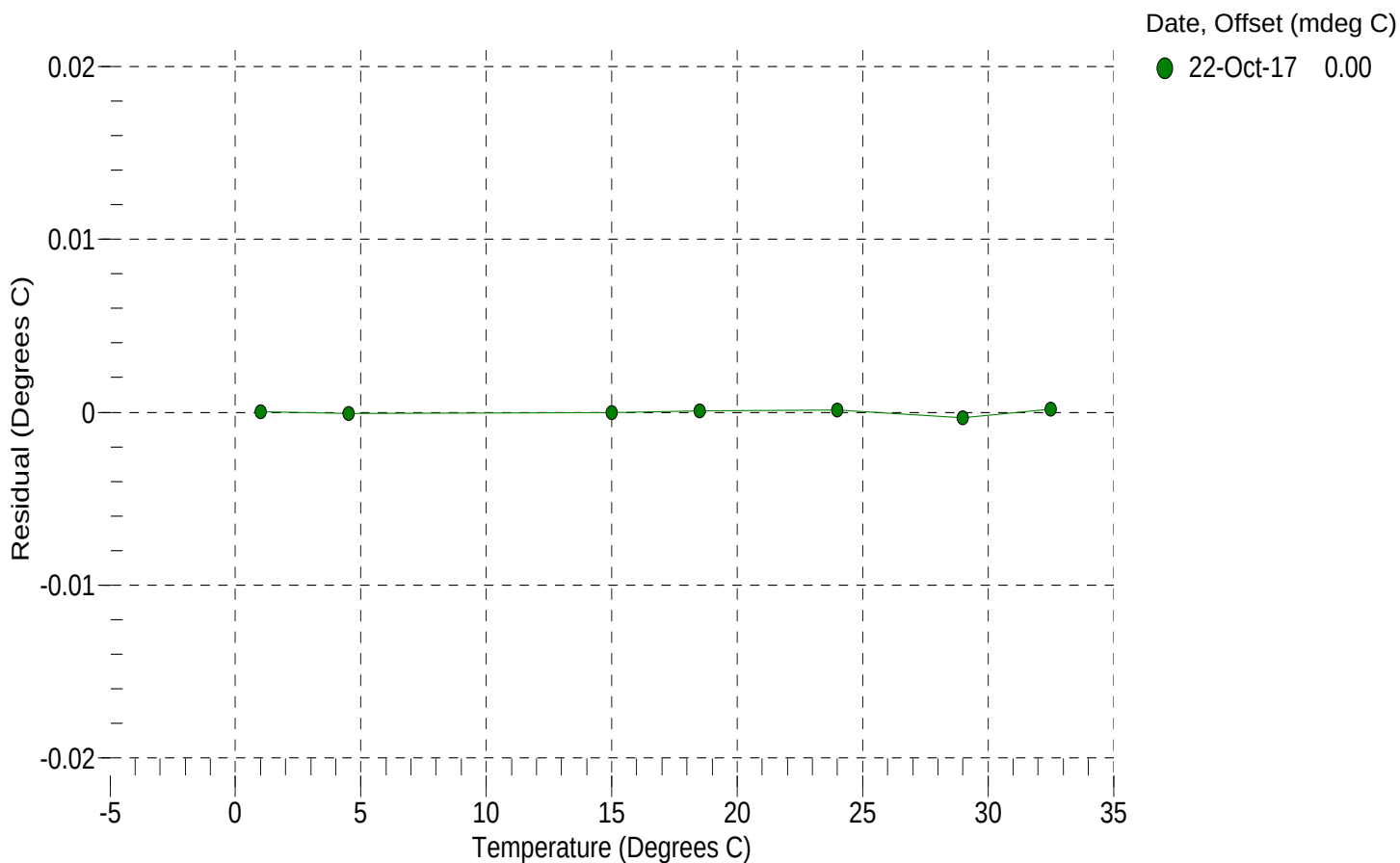
a0 = -2.423411e-004
a1 = 3.268645e-004
a2 = -5.523974e-006
a3 = 2.294039e-007

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
1.0000	559065.8	1.0000	0.0000
4.5000	479781.0	4.4999	-0.0001
15.0000	309153.9	15.0000	-0.0000
18.5000	268670.5	18.5001	0.0001
24.0000	216768.6	24.0001	0.0001
29.0000	179419.9	28.9997	-0.0003
32.5000	157688.0	32.5002	0.0002

n = Instrument Output (counts)

Temperature ITS-90 (°C) = $1/\{a_0 + a_1[\ln(n)] + a_2[\ln^2(n)] + a_3[\ln^3(n)]\} - 273.15$

Residual (°C) = instrument temperature - bath temperature





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SBE 37 V2 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

g = -9.850422e-001
h = 1.228657e-001
i = -4.473006e-005
j = 1.853253e-005

CPcor = -9.5700e-008
CTcor = 3.2500e-006
WBOTC = -5.8863e-008

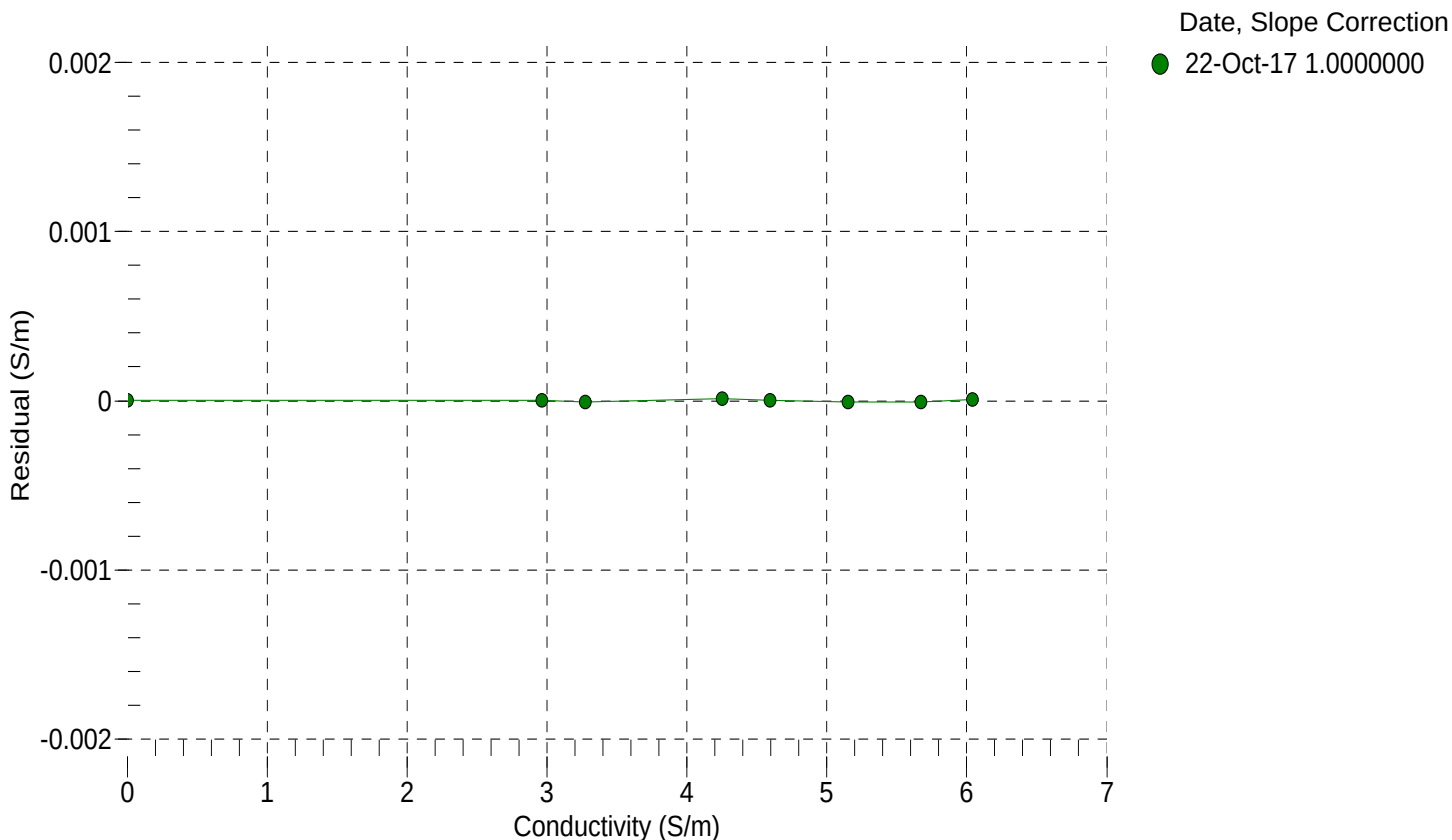
BATH TEMP (° C)	BATH SAL (PSU)	BATH COND (S/m)	INSTRUMENT OUTPUT (Hz)	INSTRUMENT COND (S/m)	RESIDUAL (S/m)
22.0000	0.0000	0.00000	2831.22	0.00000	0.00000
1.0000	34.6826	2.96562	5662.64	2.96562	0.00000
4.5000	34.6632	3.27170	5877.06	3.27169	-0.00001
15.0000	34.6221	4.25030	6514.68	4.25031	0.00001
18.5000	34.6140	4.59442	6724.26	4.59442	0.00000
24.0000	34.6052	5.15068	7049.59	5.15068	-0.00001
29.0000	34.6009	5.67100	7340.55	5.67099	-0.00001
32.5000	34.5986	6.04231	7541.15	6.04231	0.00001

$f = \text{Instrument Output(Hz)} * \sqrt{1.0 + \text{WBOTC} * t} / 1000.0$

t = temperature (°C); p = pressure (decibars); $\delta = \text{CTcor}$; $\epsilon = \text{CPcor}$;

Conductivity (S/m) = $(g + h * f^2 + i * f^3 + j * f^4) / (1 + \delta * t + \epsilon * p)$

Residual (Siemens/meter) = instrument conductivity - bath conductivity





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SENSOR SERIAL NUMBER: 16238
CALIBRATION DATE: 09-Oct-17

SBE 37 V2 PRESSURE CALIBRATION DATA
2900 psia S/N 10755345

COEFFICIENTS:

PA0 = 4.072913e-001
PA1 = 9.060707e-003
PA2 = -1.577277e-010
PTempa0 = -5.864851e+001
PTempa1 = 5.434319e-002
PTempa2 = -6.559818e-007

PTCA0 = 5.238980e+005
PTCA1 = 1.360825e+000
PTCA2 = 3.626755e-002
PTCB0 = 2.523532e+001
PTCB1 = 6.766917e-004
PTCB2 = 0.000000e+000

PRESSURE SPAN CALIBRATION

THERMAL CORRECTION

PRESSURE (PSIA)	INSTRUMENT OUTPUT (counts)	THERMISTOR OUTPUT (volts)	COMPUTED PRESSURE (PSIA)	RESIDUAL (%FSR)	TEMP (°C)	THERMISTOR OUTPUT (volts)	INSTRUMENT OUTPUT (counts)
14.74	525533.5	1511.5	14.79	0.00	32.50	1713	525611.10
592.06	589351.3	1514.2	592.00	-0.00	29.00	1646	525602.49
1169.03	653279.6	1515.3	1168.93	-0.00	24.00	1550	525587.73
1746.13	717375.4	1516.6	1746.07	-0.00	18.50	1445	525568.48
2323.15	781609.6	1517.5	2323.16	0.00	15.00	1378	525556.10
2900.24	845979.4	1518.6	2900.17	-0.00	4.50	1179	525538.15
2323.13	781614.4	1517.9	2323.20	0.00	1.00	1113	525532.80
1746.18	717404.1	1517.1	1746.33	0.01	TEMPERATURE (°C) SPAN (mV)		
1169.10	653301.4	1517.0	1169.12	0.00			
591.79	589322.9	1517.3	591.74	-0.00			
14.73	525532.1	1518.1	14.77	0.00	-4.90	25.23	
					35.00	25.26	

y = thermistor output (counts)

t = PTempa0 + PTempa1 * y + PTempa2 * y²

x = instrument output - PTCA0 - PTCA1 * t - PTCA2 * t²

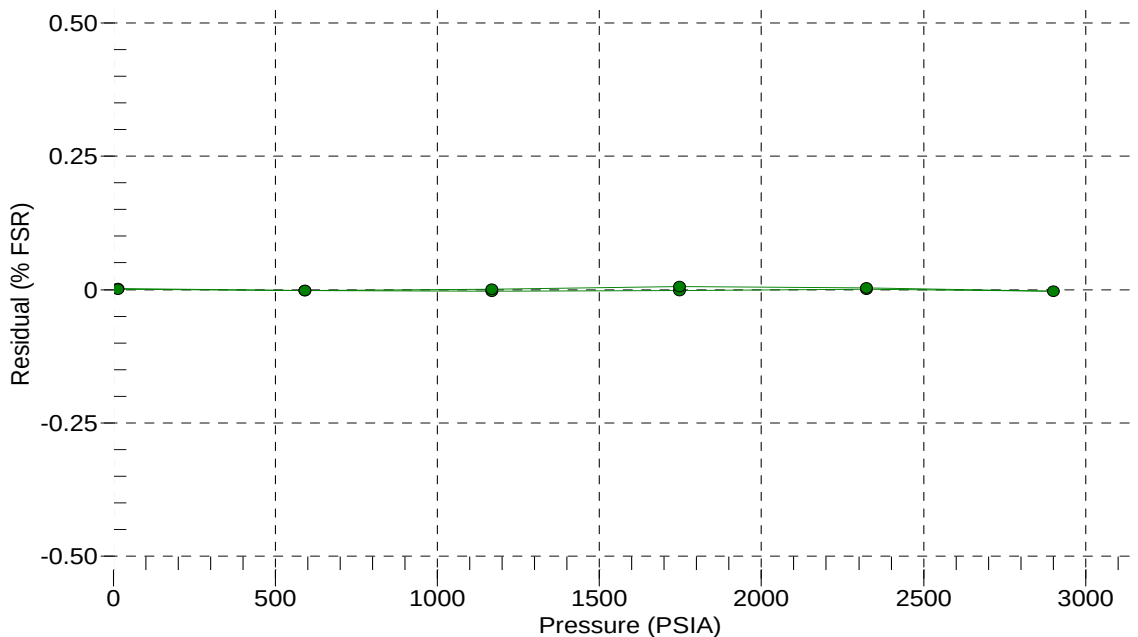
n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t²)

pressure (PSIA) = PA0 + PA1 * n + PA2 * n²

Residual (%FSR) = (computed pressure - true pressure) * 100 / Full Scale Range

Date, Offset (%FSR)

● 09-Oct-17 0.00





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Pressure Test Certificate

Test Date: **2017-10-18**

Description: **SBE-37 Microcat**

Sensor Information:

Model Number: **SBE-37**

Serial Number: **16238**

Pressure Test Protocol:

Low Pressure Test: **40** PSI Held For: **15** Minutes

High Pressure Test: **2900** PSI Held For: **15** Minutes

Passed Test: **True**

Tested By: **JY**

